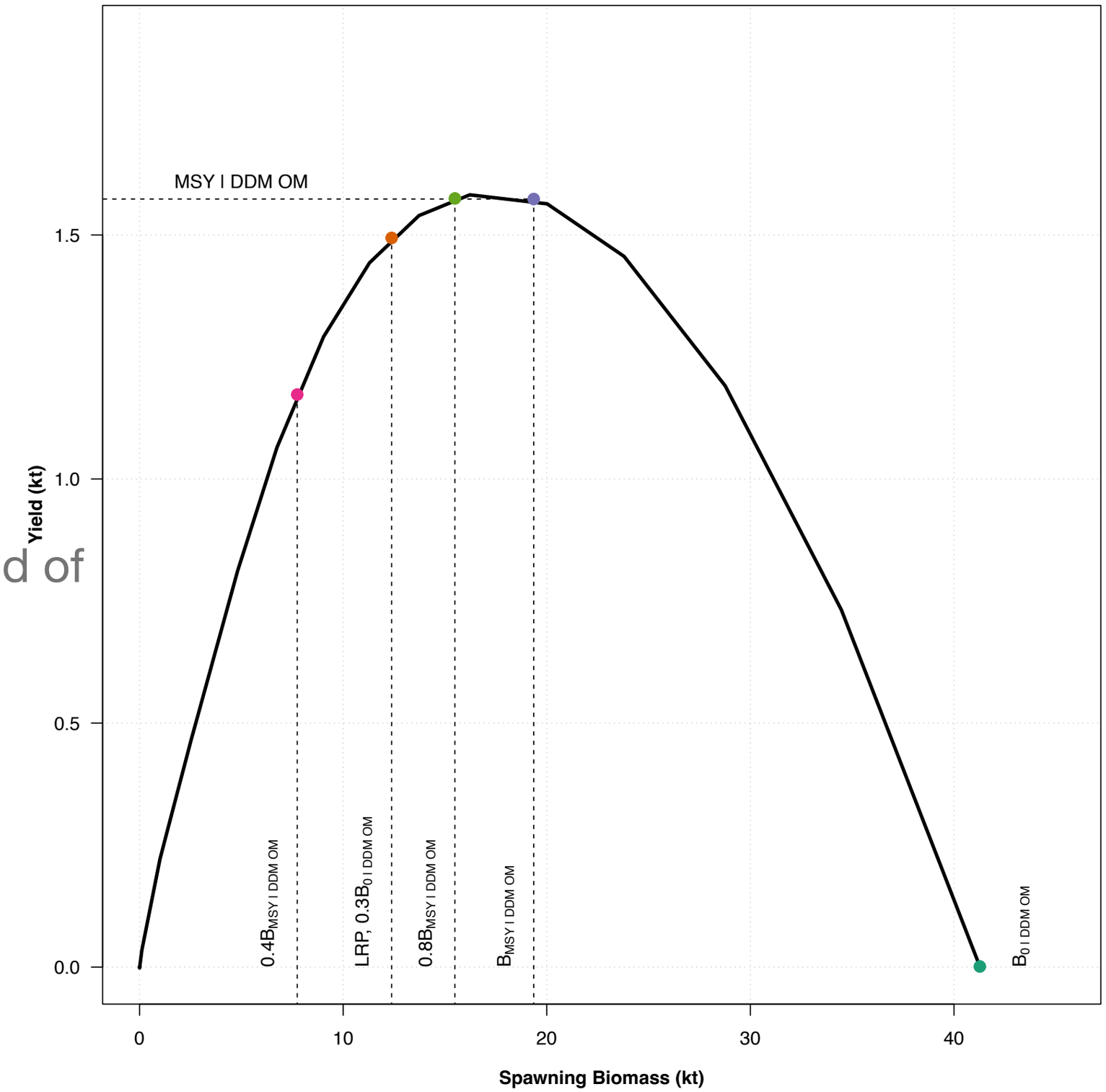
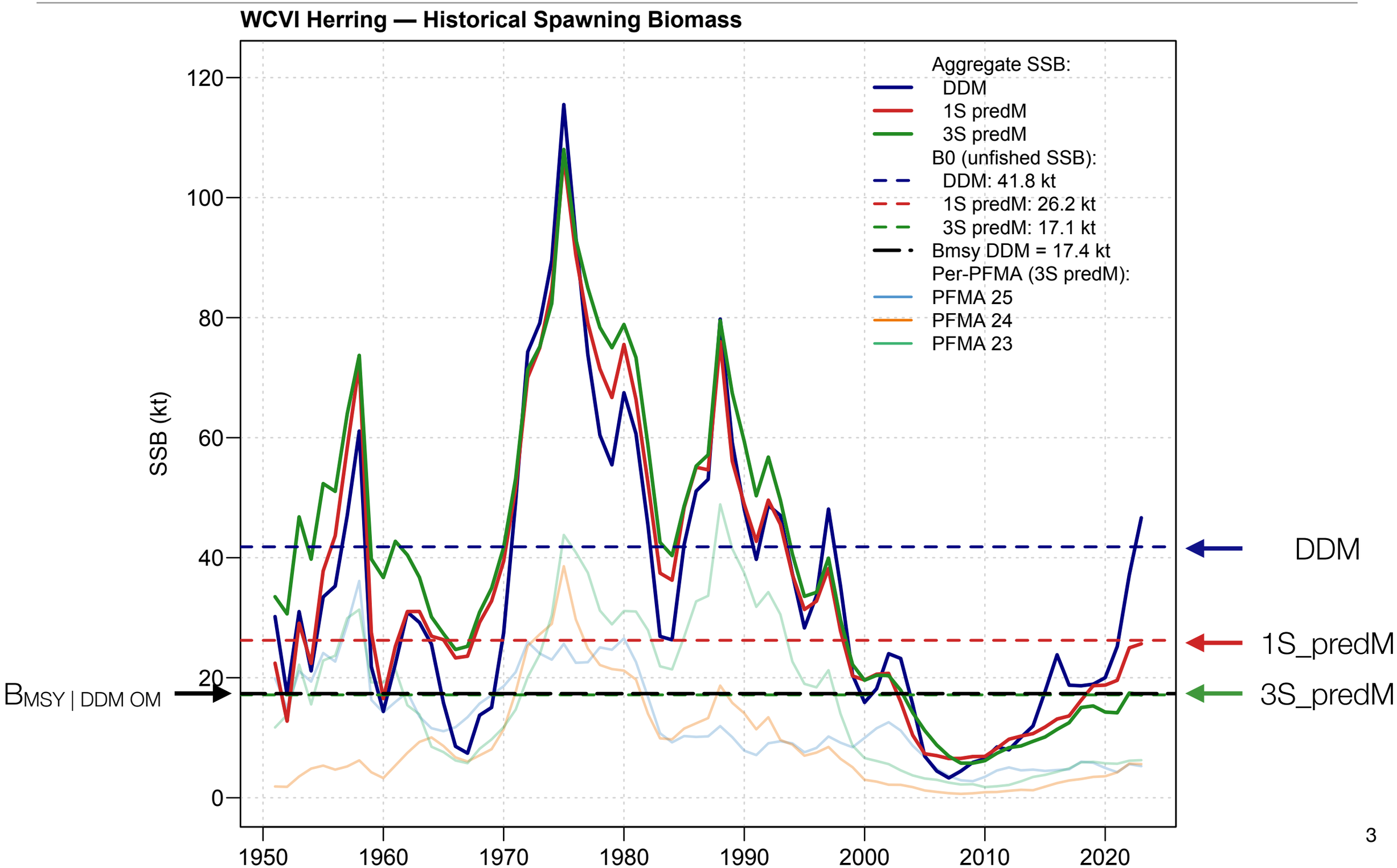
Testing a potential WCVI herring re-opening procedure under an ecosystem approach

2023: SISCAH-DDM produces WCVI herring yield curve based on 2022 data

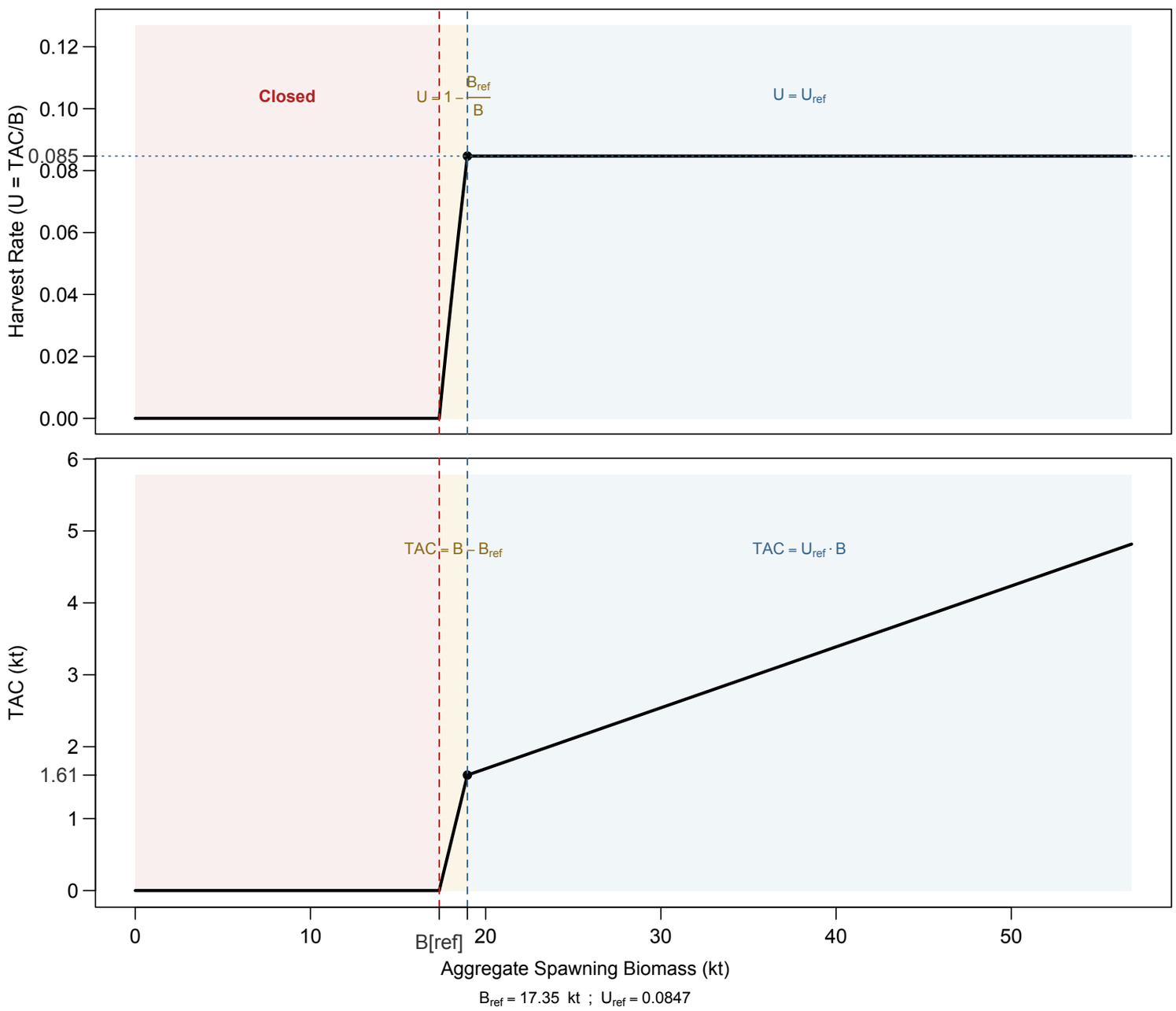
- Unfished biomass around 42,000 t
- Bmsy = 17,350 t
- LRP = 13,500 t
- Maximum sustainable yield of 1,600 t
- LRP and Bmsy are close



Uu-ath-luuk developed predation models for herring, with similar histories but different outlooks



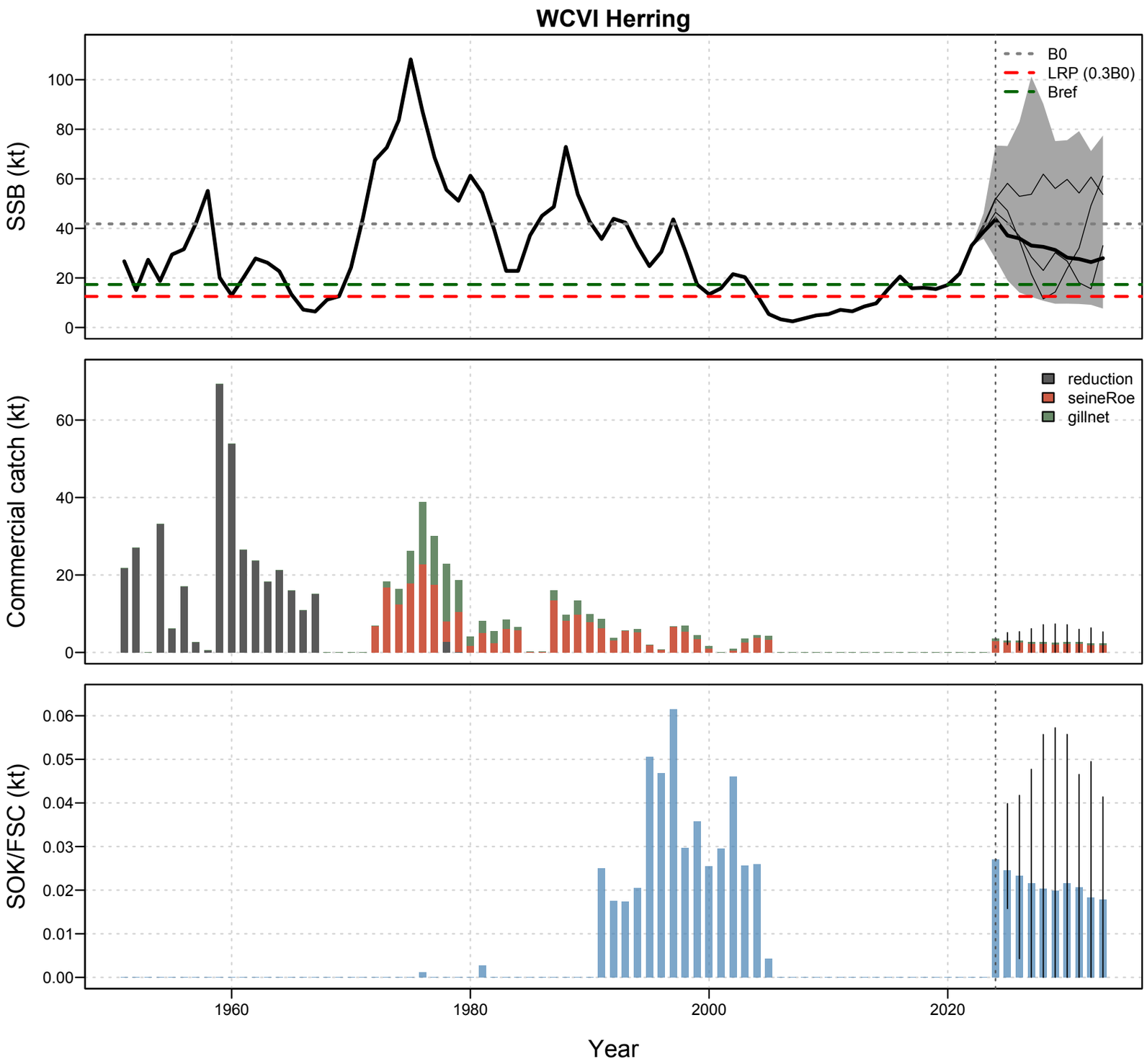
Proposed minimum escapement rule, similar to original rule



- Uses a minimum escapement of Bmsy DDM = 17,350 t
- Allows for up to an 8.5% harvest rate (blue region), as long as minimum escapement is respected (yellow region)
- Yield at lower end of blue region about MSY

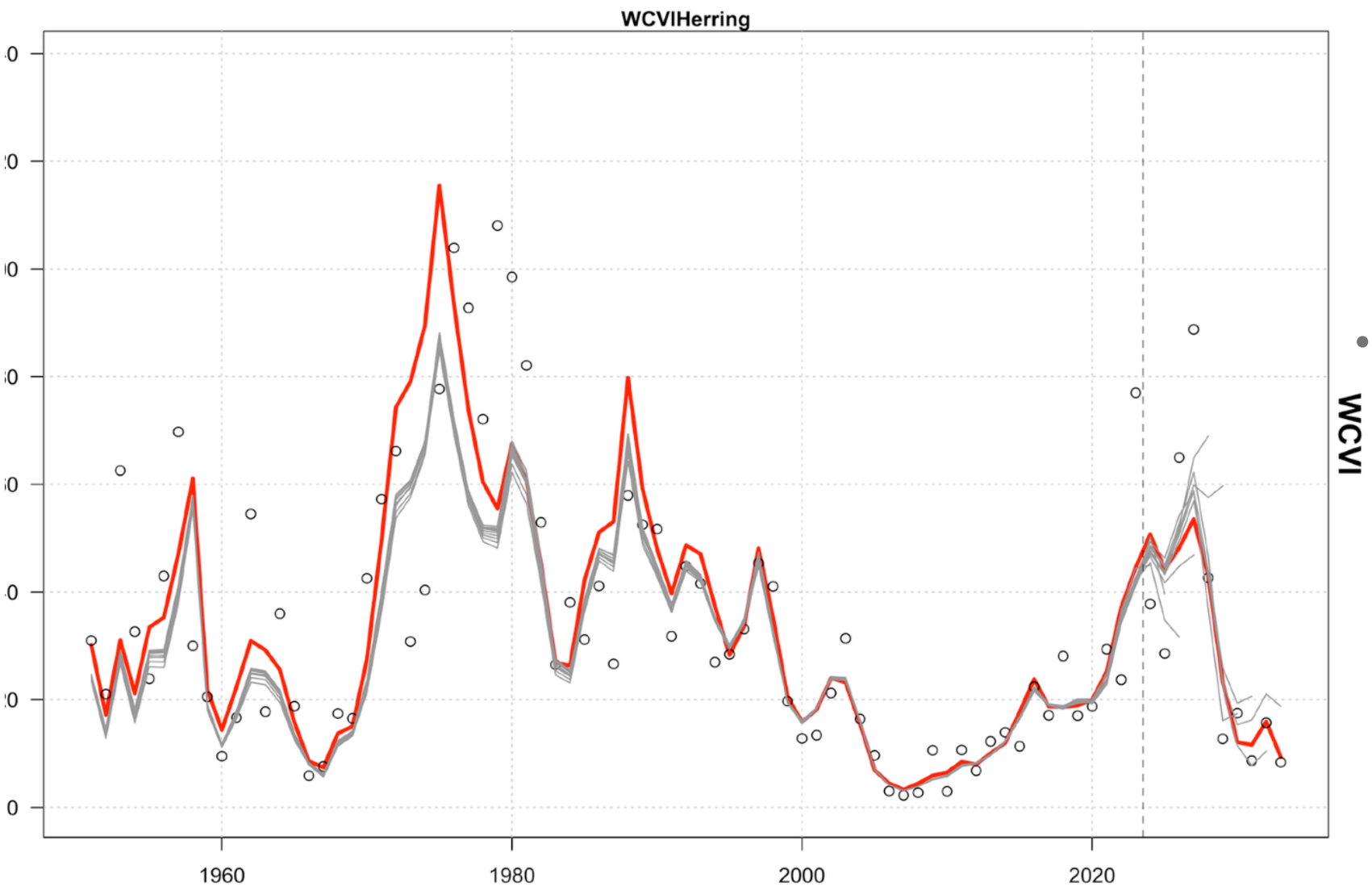
Applying MP to DDM OM

Testing the rule under the DDM 1-stock OM - no surprises, given tuning



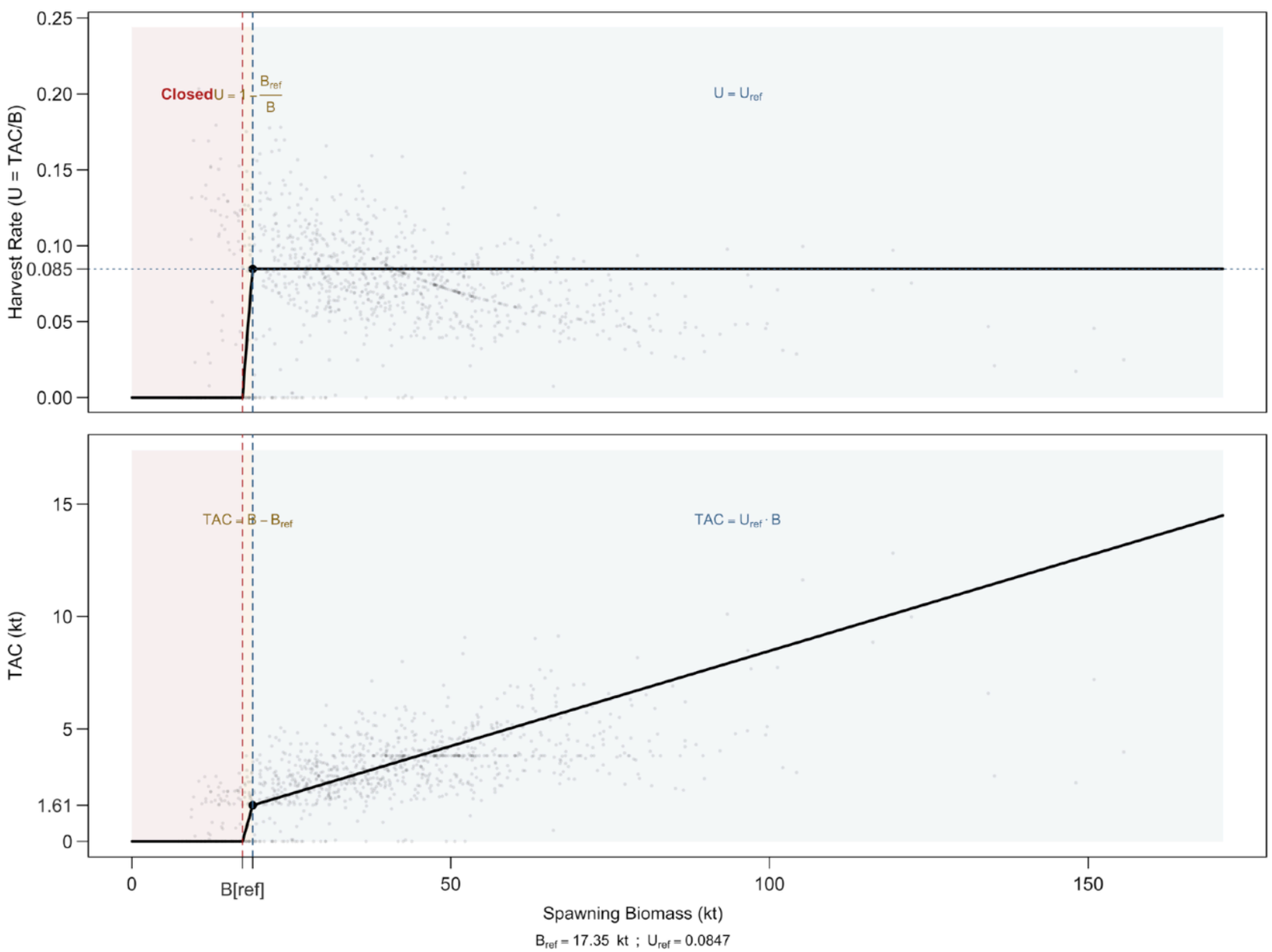
- Expected level of catch in projection, split among roe fleets
- SOK catches similar average level to historical period
- Biomass above LRP 75% of the time, and median biomass well above $B_{ref} = B_{MSY|DDM\ OM}$ level

Effective harvest rates compare well enough to the rule as defined



- SISCAH DIM model does a fairly decent job of estimating DDM OM biomass
- No major evidence of bias in projection years

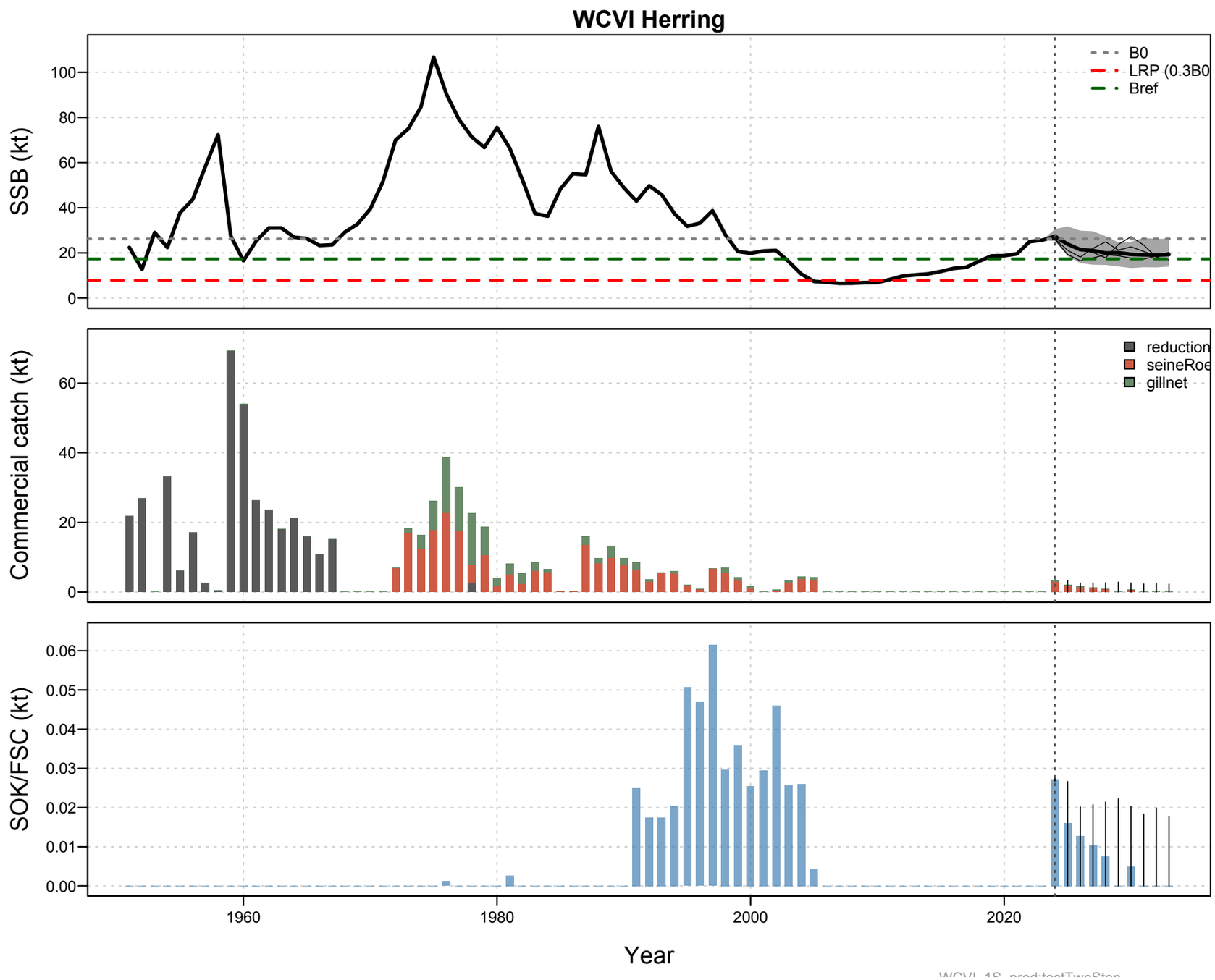
Effective harvest rates compare well enough to the rule as defined



- Line of points indicating the first year TAC
- Some over-harvesting around the minimum escapement level, indicating over-estimates of biomass by the SISCAH-based MP

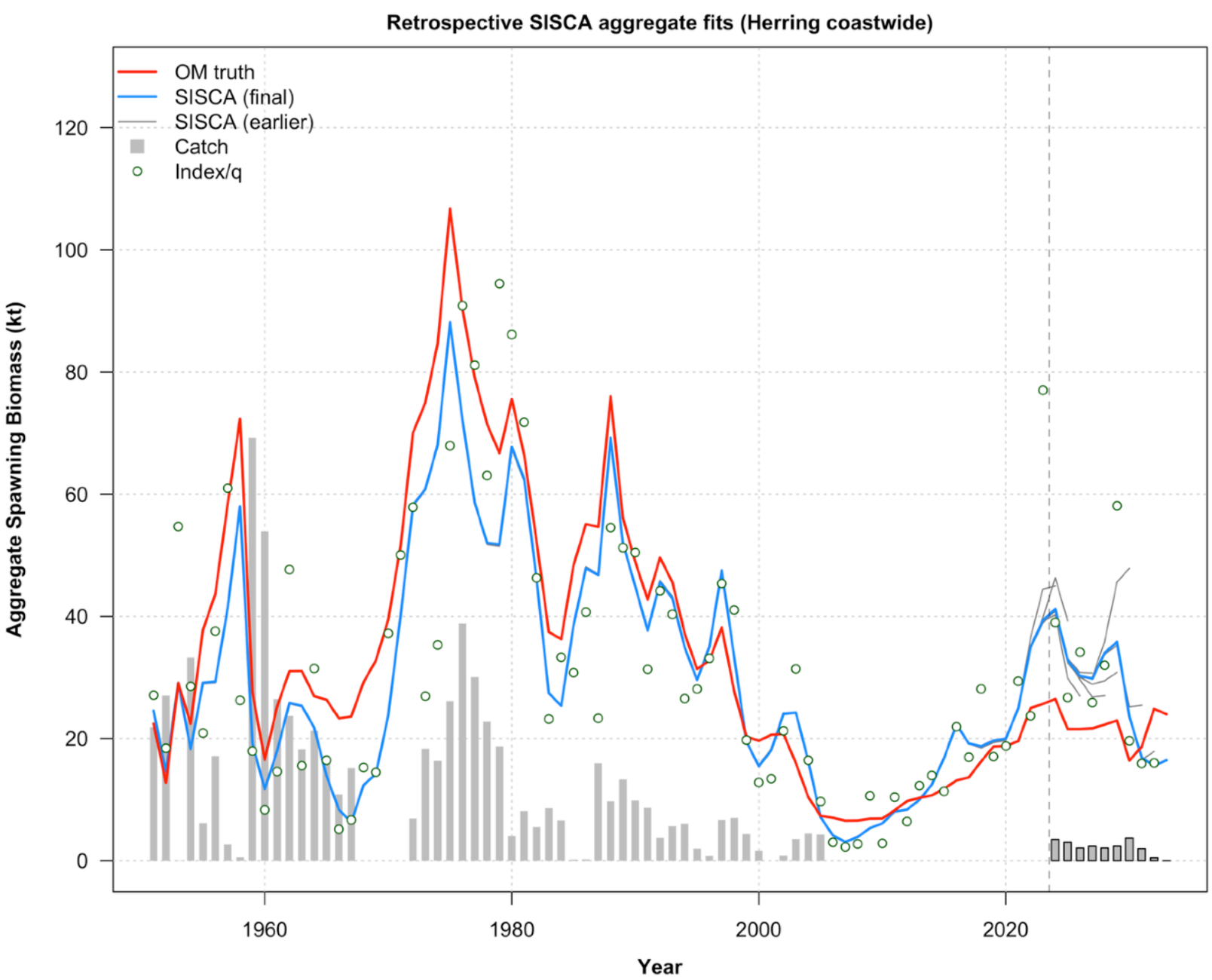
Applying MP to 1S-predM OM

Testing the rule under the predM 1-stock OM - lower catches given lower recent/future biomass



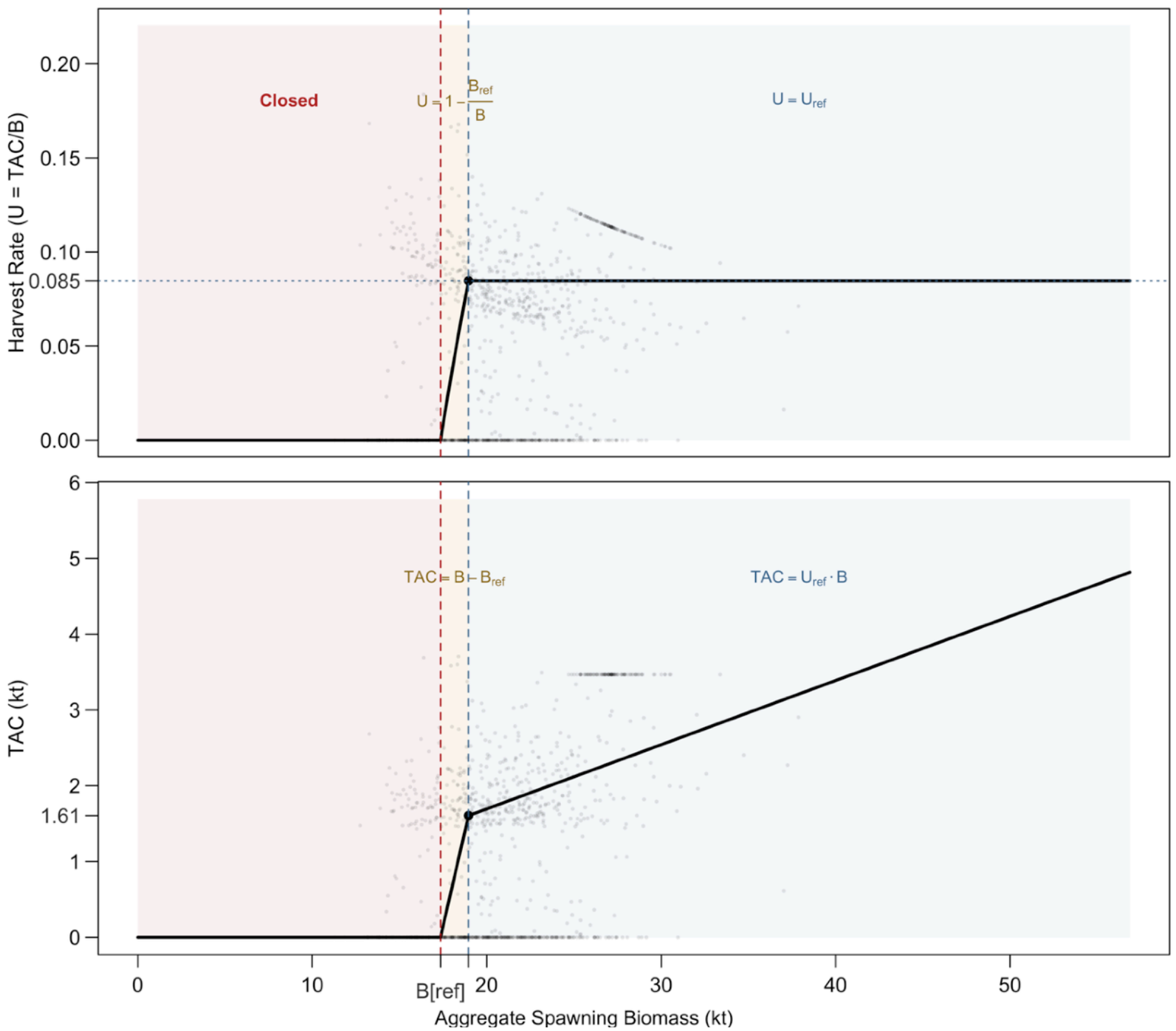
- Reduced catch in projection period across roe fisheries and FSC
- Biomass starts out lower, and declines to around the $B_{ref} = B_{MSY|DDM\ OM}$
- Large indices in recent years drive early catches higher

Effective harvest rates compare well enough to the rule as defined



- DIM model mis-specification more problematic than under DDM OM
- As before, DIM formulation has more flexibility to chase high data points, leading to over-estimates for most of the projection period

Effective harvest rates compare well enough to the rule as defined



- Line of points indicating the first year TAC of around 3.5 kt
- As before, some evidence of assessment error based catches in the closed portion of the rule
- Lots of points show closures when B > Bref as well (more than DDM OM)

Applying MP to 3S-predM OM

Testing under the 3S model - allocation methods

- To test under the 3S model, we need to decide how to allocate to each of the three smaller areas
- We used three methods:
  - Recent average spawn indices
  - (Most) recent average catch
  - 100% PFMA 23 (Barkley Sound)

2024 Allocation

	SI	Catch	All 23
PFMA 25	5%	30%	0%
PFMA 24	38%	32%	0%
PFMA 23	57%	38%	100%

Catch allocation based on 2002 - 2004 in 2024

Testing under the 3S model - allocation methods

- To test under the 3S model, we need to decide how to allocate to each of the three smaller areas
- We used three methods:
  - Recent average spawn indices
  - (Most) recent average catch
  - 100% PFMA 23 (Barkley Sound)

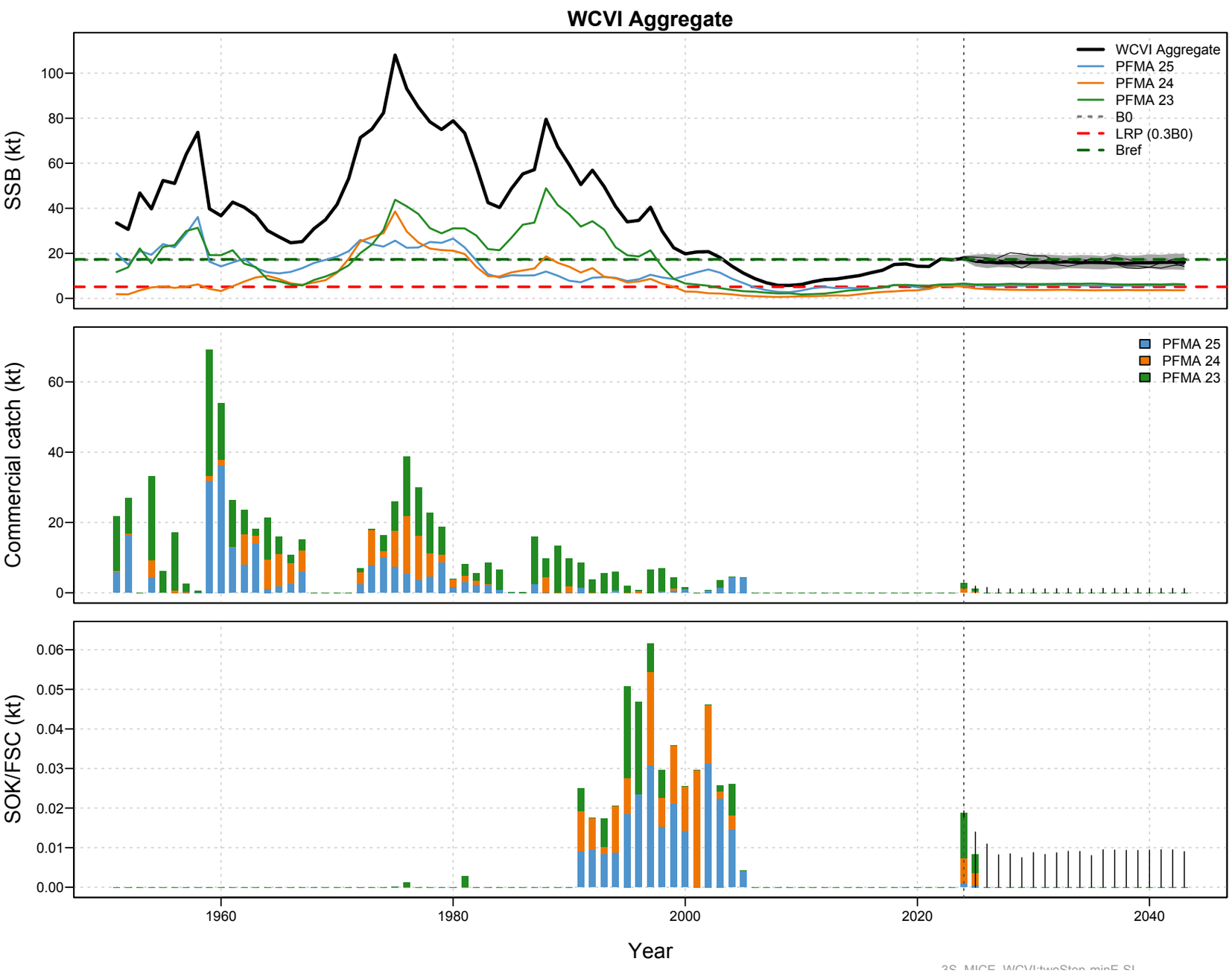
Average Allocation

	SI	Catch	All 23
PFMA 25	11%	25%	0%
PFMA 24	31%	28%	0%
PFMA 23	58%	47%	100%

Evaluating performance under 3S model

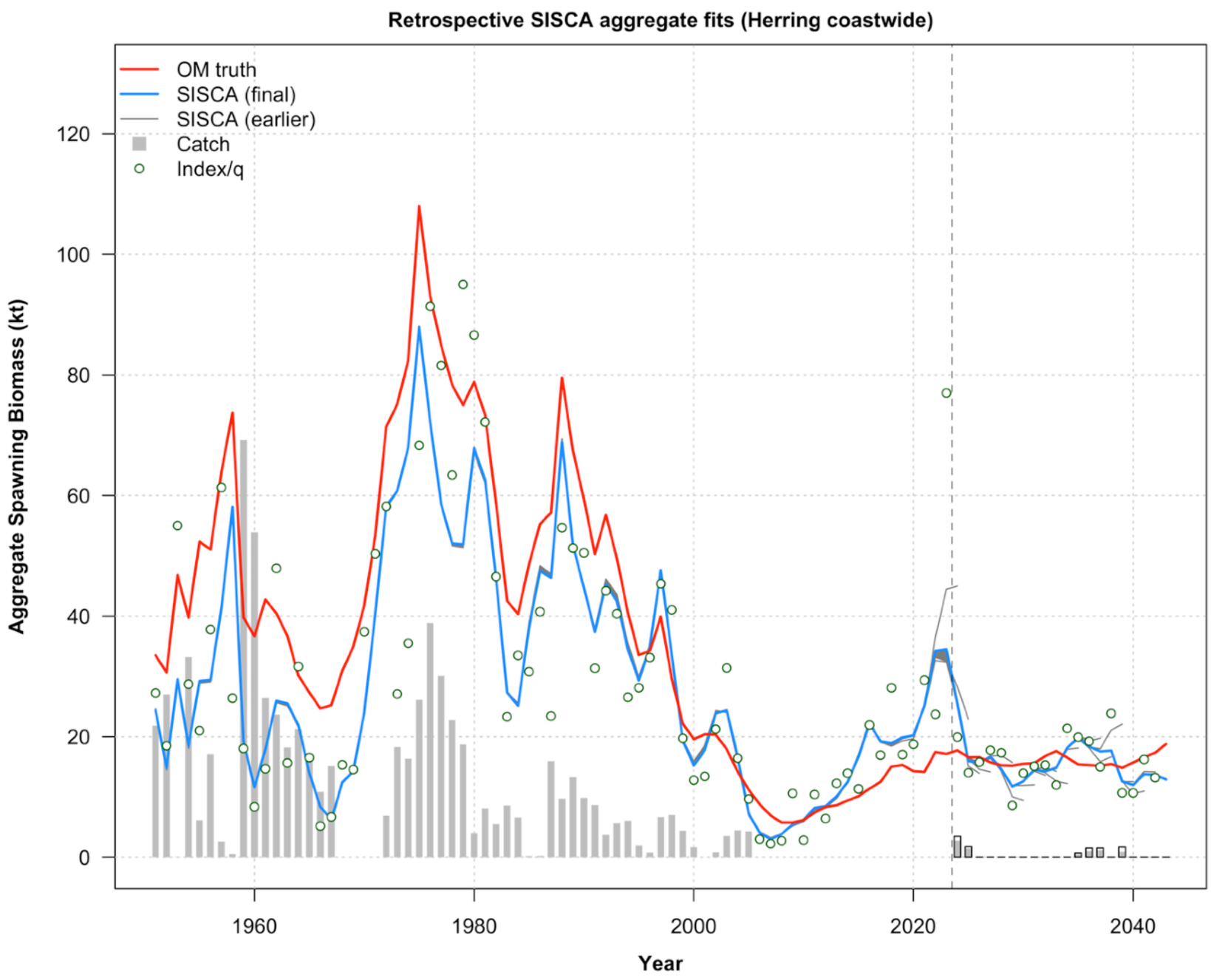
- No reference points at the PFMA level, so we define:
  - Poor biomass: probability that biomass is below the historical minimum level
  - Poor catch: the probability that catches are below the historical minimum level

Testing the rule under the predM 3-stock OM - catches are lower again



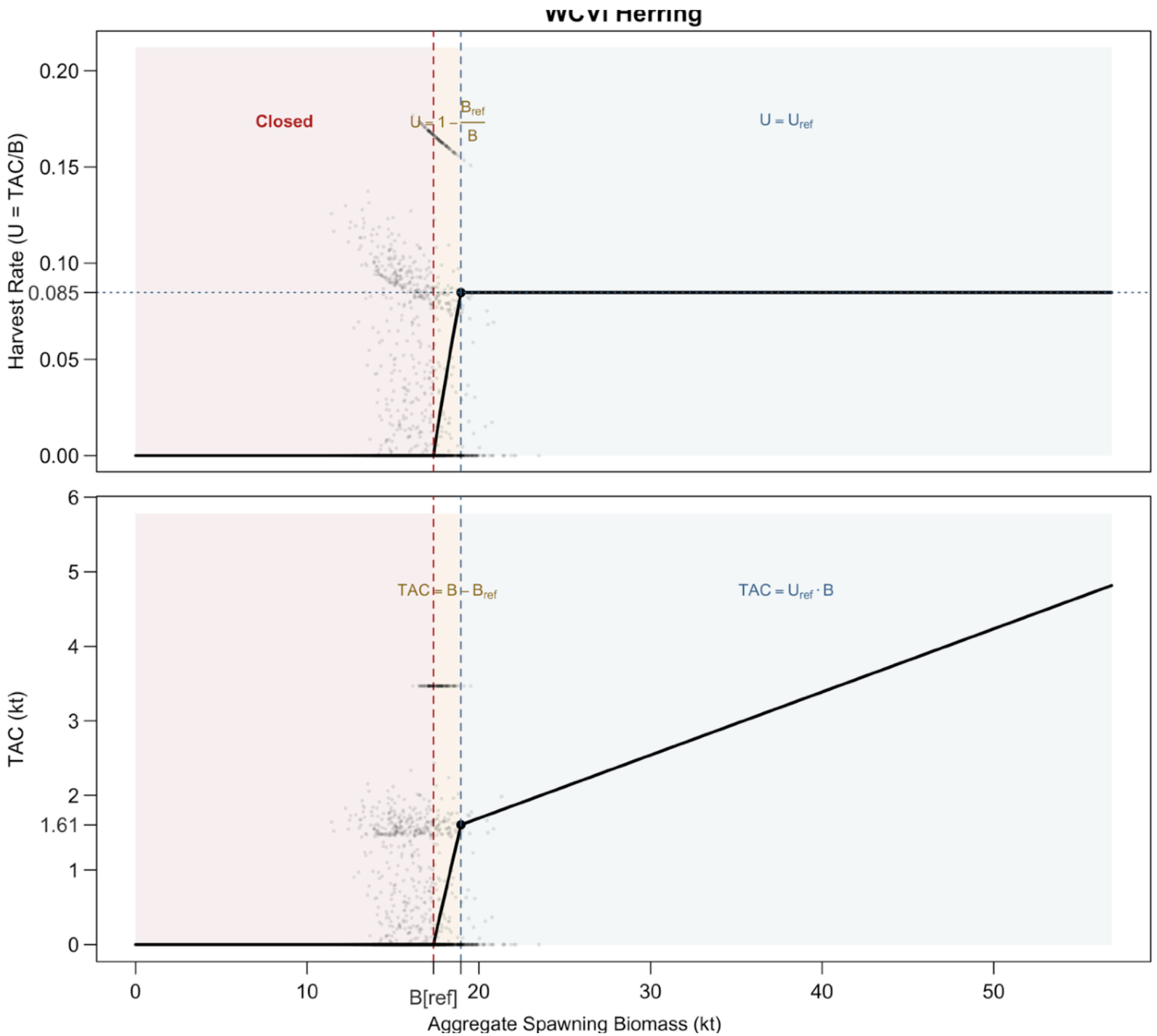
- Over 50% probability of zero catch, as biomass is already around the B_{ref} = $B_{MSY|DDM\ OM}$ level
- Largest catches in first 2 years given recent high spawn index points

Effective harvest rates compare well enough to the rule as defined



- As before, DIM formulation has more flexibility to chase high data points, leading to over-estimates for most of the projection period
- Flat biomass and indices in projection period reduce scale of overestimates more quickly

Effective harvest rates compare well enough to the rule as defined



- Line of points indicating the first year TAC of around 3.5 kt
- Assessment errors are more of an issue now, as biomass is flatlining around the OM's unfished biomass level, where the fishery is closed anyway

Allocation has very little effect, given low catches in simulations

- Catches are almost zero, so biomass performance is largely driven by natural/predation mortality and recruitment

- Very little difference among allocation methods or areas (if catch is taken)

Probability of Low Biomass

	SI	Catch	All 23
PFMA 25	0.2%	0.2	0.2
PFMA 24	0%	0%	0%
PFMA 23	0%	0%	0%

Allocation has very little effect, given low catches in simulations

- Catches are almost zero, so biomass performance is largely driven by natural/predation mortality and recruitment
- Very little difference among allocation methods or areas (if catch is taken)

Probability of Low Catch

	SI	Catch	All 23
PFMA 25	71%	71%	100%
PFMA 24	72%	72%	100%
PFMA 23	71%	71%	72%